

Earthline Protocol Design Document

Elevator Pitch Ideas

- _____ is a climate change themed tower defense game where you use real-world solutions to defend the planet from natural/unnatural disasters.

Pillars

- Serious situations, Approachable Aesthetic
- Use real solutions/situations where applicable

Game Systems

- Round/Wave-based combat: The player engages with only one wave of enemies within a given round.
 - Each game is 30 rounds.
- Health-based Units instead of the bloons way.
 - Bloons Tower Defense uses a shell system where a parent balloon will spawn smaller balloons and the only ones that require multiple hits are from ceramic upwards. The problem with the theme of our game is its hard to find a natural parental curve between all of the natural/unnatural disasters.
- Enemies as scriptable objects/prefabs
 - All enemies should be born from a parent class and have a shared set of information that can be tweaked on an individual basis between each tower.
 - Example of the serialized fields:
 - Health
 - Movement Speed
 - Size (Radius)
 - Resistant to X
 - Vulnerable to X
- Towers as scriptable objects/prefabs
 - Similar to enemies, towers should be born of a parent class and be able to perform actions like “Attack targets within range” and/or “Target the first enemy on the road within range”. Only exception would be for towers that don’t shoot directly at the enemies and instead buff allies or place traps on the road.

- NASA Satellites: Shows the player what the next round of disasters will include, allowing them to strategize and respond to the situation
- Global Temperature and City Health as Win/Loss conditions

Global Temperature

- Measured in Fahrenheit
 - Optional Feature: Adding a toggle to switch to Celsius
- Starts at 56 degrees
- Increases by .2 per round played
- Game Over at 60 degrees (20 rounds normal gameplay without installing any improvements)

City Health

- Base Health: 200
- Game over at 0 health.

Towers

For now, towers have 5 variable stats; Cost, Damage, Attack Speed, Range (we will figure out a unit of measurement next week), and “Favored Disaster(s)” (Work in progress name).

Stats Breakdown

- Cost: Dictates how much money is lost when purchasing a tower
- Damage: How much damage **each projectile** does when hitting an enemy.
- Attack Speed: How frequently a tower shoots each projectile.
 - Should be either shots per second or delay per shot
 - Ex. 4 per second = .25 delay per shot
- Range: How large the attack radius of the tower is in some sort of standardized unit.
- Favored Disaster(s): What disasters a tower will deal **Bonus Damage** to
 - For now, bonus damage will be Damage x 1.5
 - For ease of implementation, it would be awesome if it could be an array/serialized field that we could just drag and drop enemy prefabs into.
 - If that’s not how this works, forgive me

Tower Ideas

Sorted by most likely to be included

Winds-Anti-Shear-Propulsor (W.A.S.P)

- **Description:** Counters Tornadoes and Hurricanes by disrupting their wind source
- **Cost:** 450
- **Damage:** 12
- **Attack Speed:** 2 per second
- **Range:** 200
- **Favored Disaster(s):** Hurricanes, Tornadoes

Flying Planter (Name is a W.I.P)

- **Description:** Counters landslides by making the soil more stable and revitalizes dying habitats. Bird that drops seeds on the ground.
- **Cost:** 300
- **Damage:** 20
- **Attack Speed:** 1 per second
- **Range:** 7
- **Favored Disaster(s):** Landslide

Saltwater Purifier (Name is a W.I.P)

- **Description:** Counters droughts and wildfires by purifying saltwater and blasting it at enemies. Fish in a suit with a bubble on its head like Sandy from Spongebob
- **Cost:** 600
- **Damage:** 5
- **Attack Speed:** 5 per second
- **Range:** 9
- **Favored Disaster(s):** Wildfire, Volcano

Carbon Capture (Name is a W.I.P)

- **Description:** Counters wildfires and volcanoes. Bear that sucks it up like a vacuum
- **Cost:** 700
- **Damage:** 1
- **Attack Speed:** 30 per second
- **Range:** 7
- **Favored Disaster(s):** Heatwave, Wildfire

Polar Air Solutions (Heating/Cooling Solution)

- **Description:** Polar bear that chills enemies with an electric air pump. Kind of like the tack shooter in Bloons Tower Defense (Short AOE around tower)
- **Cost:** 700
- **Damage:** 3
- **Attack Speed:** 10 per second
- **Range:** 4.5
- **Favored Disaster(s):** Heatwave, Volcano

Barricading Beaver

- **Description:** Beaver that creates dams to block Landslides and Floods.
- **Cost:** 500
- **Damage:** 12
- **Attack Speed:** 2 per second
- **Range:** 9
- **Favored Disaster(s):** Landslide, Floods

In-City Upgrades and Opposition

The goal of in-city upgrades is to display what kind of social/personal/governmental decisions can have an impact on climate. It should also provide a secondary challenge from the player beyond the natural disasters they are facing.

Player Controlled (One time purchases)

- **Energy**
 - Solar Photovoltaic Field
 - Cost: 300
 - Unlock Round: 5
 - Effect: Reduces Global Temperature increase by .01
 - Wind Turbine Farm
 - Cost: 600
 - Unlock Round: 10
 - Effect: Reduces Global Temperature increase by .01
 - **Hydroelectric Dam**
 - Cost: 900
 - Unlock Round: 15
 - Effect: Reduces Global Temperature increase by .02

- Solar Concentrated Power
 - Cost: 1200
 - Unlock Round: 20
 - Effect: Reduces Global Temperature increase by .02
- Geothermal
 - Cost: 1500
 - Unlock Round: 25
 - Reduces Global Temperature increase by .02
- **Efficiency**
 - Public Transportation
 - Cost: 300
 - Unlock Round: 15
 - Reduces Global temperature increase by .01
- **~~Social (SCRAPPED)~~**
 - ~~Fact-checking outlet~~
 - ~~Cost:~~
 - ~~Unlock Round: 17~~
 - ~~Effect: Corrects conspiracy theorists, reducing the harm of disinformation by 80%~~
 - ~~Environmental Protests~~
 - ~~Cost:~~
 - ~~Unlock Round:~~
 - ~~Effect: Urges politicians to care about the environment, increasing all money earned by 10%~~

~~Oppositional Events (SCRAPPED)~~

- **~~Conspiracy Theorists~~**
 - ~~Start Round: 15~~
 - ~~Increases Health of all Natural Disasters (makes them harder to deal with)~~
- **~~Corrupt Politician~~**
 - ~~Start Round: 10~~
 - ~~Decreases money earned (idea that political climate action can be hindered by disinformation)~~

Enemies

Towers have 4 variable stats and a “Special Effect”; Health, City Damage, Movement Speed (Scale of 1-10), and Size (Diameter).

Enemy Ideas

- Tornadoes
 - **Health: 50**
 - **City Damage: 2**
 - **Movement Speed: 6**
 - **Size: .5**
 - **Special Effect:** N/A
 - **Money Awarded: 30**
 - Art: Would be really cool if they always spun counter-clockwise
- Hurricanes
 - **Health: 150**
 - **City Damage: 1**
 - **Movement Speed: 4**
 - **Size: 1**
 - **Special Effect:** Spawns 2 tornadoes upon death.
 - **Money Awarded: 30**
 - Art: Should be quite a bit larger than a tornado
 - **Real World Solutions:** Mostly resilience and infrastructure that allows a timely response for victims
- Heatwave
 - **Health: 250**
 - **City Damage: 1**
 - **Movement Speed: 2**
 - **Size: 1**
 - **Special Effect:** Causes towers in the area to shoot 20% slower
 - **Money Awarded: 50**
 - Real World Solutions: *Public air conditioned places, planting trees/vegetation (absorbs the carbon out of the air)*
- Landslide
 - **Health: 10 (Low), 25 (Medium), 50 (Large)**
 - **City Damage: 1**
 - **Movement Speed: 5**
 - **Size: .1 (Low), .2 (Medium), .5 (Large)**
 - **Money Awarded: 50**

- **Special Effect:** 6 low health rocks, 4 medium health rocks, and 2 large health rocks
- **Wildfires**
 - **Health:** 200
 - **City Damage:** 5 base + 1 every 2 seconds
 - **Movement Speed:** 4
 - **Size:** .5 + 1 every 2 seconds
 - **Money Awarded:** 50
 - **Special Effect:** Every 2 seconds, increases size and city damage by 1
- **Floods:**
 - **Health:** 200
 - **City Damage:** 5
 - **Movement Speed:** 6
 - **Size:** 1
 - **Money Awarded:** 50
 - **Special Effect:** Causes towers in the area to deal 10% less damage
- **Boss Idea: Volcanic Eruption**
 - **Health:** 1000
 - **City Damage:** 30
 - **Movement Speed:** 2
 - **Size:** 1
 - **Money Awarded:** 200
 - **Special Effect:** Slow and deadly 😊
- **Boss Idea: Tsunami**
 - **Health:** 800
 - **City Damage:** 30
 - **Movement Speed:** 6
 - **Size:** 1
 - **Money Awarded:** 200
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Map

